

SOBHAN KASAEI

🌐 [SobhanKasaei.github.io](https://github.com/SobhanKasaei) | ✉ Sobhan.Kasa@gmail.com | [🌐 LinkedIn](#) | [🐙 GitHub](#)

Education

Sharif University of Technology

Bachelor of Science in Industrial Engineering

Oct 2021 – Present

Tehran, Iran

- Current CGPA: 18.96/20 (3.89/4.0)
- Last Year GPA: 19.74/20 (4.0/4.0)

Sharif University of Technology

Minor in Economics

Aug 2023 – Present

Tehran, Iran

- Current CGPA: 19.78/20 (4/4)

NODET (Sampad)

Diploma in Physics & Mathematics

Jun 2018 – Jun 2021

Tehran, Iran

- CGPA: 18.86/20

Experience

Quera Company

Mentor — Artificial Intelligence Bootcamp

Jul 2024 – Feb 2025

Tehran, Iran

- Mentored two teams of four students each through weekly sessions, providing guidance on core concepts in machine learning and deep learning, as well as assistance with assignments and projects.
- Designed and implemented a practical project for students to apply classical machine learning algorithms and data-cleaning techniques.

1&1 Company

International Marketing Intern

Jul 2024 – Sep 2024

Tehran, Iran

- Worked under the guidance of Prof. Majid Rafiee and Dr. Ghazimoradi.
- Assisted 1&1 in identifying export and production opportunities abroad in the agricultural sector, including potential partners and suppliers, market trend analysis, and guidance on regulations and logistics to support international expansion.
- Collected and analyzed agricultural data by scraping datasets from FAO, EFSA, and various Ministries of Agriculture.

Sharif University of Technology Specialized Training Center

Mentor — Data Science Introductory Course

Jan 2023 – Jul 2023

Tehran, Iran

- Graded assignments and provided constructive feedback to students.
- Reviewed course material with students individually and in small groups.

Notable Projects

The Relationship Between Income Inequality and Crime Rates in U.S. States

Econometrics Course Project – Lecturer: Dr. Naser Amanzadeh

2024

- Built a state-level panel of inequality and crime indicators from official U.S. sources.
- Replicated a Swedish econometric study using OLS/FEM models in STATA.
- Estimated the effect of inequality (Gini) on violent and non-violent crime with robustness checks.
- Interpreted results using standard economic theory on inequality and crime.

Analyzing the Causal Effect of Intercity Traffic on Voter Turnout

Regression Analysis Course Project – Lecturer: Dr. Kasra Alishahi

2024

- Focused on Iran's 2016 parliamentary elections to investigate abnormal voter turnout across districts.
- Integrated intercity traffic counter data with electoral statistics to identify suspicious mobility patterns.
- Developed regression models to assess whether traffic volumes influenced turnout rates.
- Interpreted results in the context of political geography and voting behavior in Iran.

Simulation of Patient Flow in a Hospital

2024

Fun of Comp Simul Course Project – Lecturer: Dr. Nafiseh Sedghi

- Developed a DES model of hospital flows including triage, labs, surgery, ICU/CCU, and wards.
- Implemented the simulation in Python with resource constraints and stochastic pathways.
- Calibrated arrivals and service times; validated behavior via tracing and stability checks.
- Evaluated key KPIs: queue lengths, waiting times, system time, and bed utilization.
- Ran sensitivity tests and compared capacity scenarios for stability improvements.

Prediction and Optimization of NYC Green Taxi Services

2024

Transportation Planning Course Project – Lecturer: Dr. Erfan Hasannayebi

- Processed multi-year NYC taxi trip data and engineered spatial-temporal features.
- Built ML models to forecast zone-level hourly demand (RF, XGBoost).
- Formulated a MILP to optimize vehicle allocation and repositioning across zones.
- Solved using Pyomo-Gurobi and evaluated improvements over baseline strategies.
- Summarized insights and operational recommendations for fleet management.

Teaching Assistantships

Engineering Statistics — Head TA

Spring 2024

Dr. Majid Khedmati & Dr. Zahra S. Hasanpour Jesri

- Designed and implemented, for the first time in the course, a project combining core course concepts with additional advanced topics; also taught these supplementary topics to students.
- Supervised a team of TAs, coordinated assignment, quiz, and exercise creation, and managed the grading and evaluation of all course components including projects.
- Organized and led weekly review and problem-solving sessions, oversaw content preparation for exercises, quizzes, and projects, and delegated responsibilities among TAs to ensure smooth course delivery.

Engineering Statistics — Head TA

Fall 2024, Spring 2025

Dr. Zahra S. Hasanpour Jesri

- Continued supervising a team of TAs, coordinating assignments, quizzes, exercises, and project design and grading.

Fundamentals of Python Programming — Head TA

Fall 2023

Eng. Alireza Kadivar

- Managed the overall course workflow, including consolidating and announcing final grades.
- Supervised TAs in grading assignments, quizzes, and exams, ensuring accurate and timely evaluation.
- Co-designed, oversaw, and coordinated project assignments; assisted in project evaluation and presentations alongside TAs.

General Economics

Fall 2024

Dr. Mohammad Vesal

- Designed and graded all course assessments, including assignments, projects, quizzes, midterm, and final exam.
- Held weekly problem-solving and review sessions to support students' understanding of course material.
- Collaborated with the instructor to ensure alignment between assessments and learning objectives.

Other Courses — Teaching Assistant

Fall 2022 – Present

Various Professors

- Data Mining Basics and its Applications, Microeconomics, Operations Research II, Statistical Quality Control, Advanced Programming, Engineering Economics.

Voluntary/Extracurricular Activities

GAMEIN Event, Sharif University

Jun 2022 – Jun 2023

Data Analysis Team Member

Tehran, Iran

- Cleaned and consolidated historical event data and maintained the data archive.
- Designed a dashboard for the closing event.

Rasta Scientific Association

Jul 2023 – Aug 2023

Instructor / Member - Optimization Workshop

Tehran, Iran

- Designed and delivered an LP workshop targeted at high-school students; taught LP fundamentals and posed exercises.

Selected Courses*

Faculty of Ind. Eng.		Faculty of Comp. Eng. / Sci.		Faculty of Econ. & Mgmt.	
Operation Research I	18.3/20	Linear Algebra	19.4/20	General Economics	20/20
Operation Research II	19.5/20	Artificial Intelligence	20/20	Microeconomics	20/20
Decision Making	18.2/20	Machine Learning	20/20	Macroeconomics	20/20
Prod and Inv Contr	19.4/20	Regress Analysis	18.3/20	Econometrics	20/20
Eng Statistics	20/20	Stochastic Processes (Grad.)	20/20	Fund of Game Theory	18.7/20
Fun of Comp Simul	18.4/20	Deep RL (Grad.)	IP	Financial Economics	IP
Transportation Planning	20/20	Convex Optimization	IP		

* IP: In Progress (courses currently being taken this term)

Skills

- **Programming & Modeling:** Python, R, GAMS, Stata, SQL, HTML/CSS, Neo4j
- **Data Visualization & Tools:** MS Power BI, Plotly, Seaborn, Matplotlib, $\text{\LaTeX}$

Languages

Persian: Native

English: Fluent (TOEFL iBT: 109; R30, L28, S24, W27)

References

Kourosh Eshghi

Professor

Department of Industrial Engineering

Sharif University of Technology

Email: eshghi@sharif.edu

Mohammad Vesal

Associate Professor

Graduate School of Management and Economics

Sharif University of Technology

Email: m.vesal@sharif.edu

Maryam Radman

Assistant Professor

Department of Industrial Engineering

Sharif University of Technology

Email: radman@sharif.edu

Zahra Sadat Hasanpour Jesri

Assistant Professor

Department of Industrial Engineering

Sharif University of Technology

Email: hasanpourjesri@sharif.edu